

Using Image Analysis to Predict the Weight of Alaskan Salmon of Different Species

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ABSTRACT: After harvesting, salmon is sorted by species, size, and quality. This is generally manually done by operators. Automation would bring repeatability, objectivity, and record-keeping capabilities to these tasks. Machine vision (MV) and image analysis have been used in sorting many agricultural products. Four salmon species were tested: pink (*Oncorhynchus gorbuscha*), red (*Oncorhynchus nerka*), silver (*Oncorhynchus kisutch*), and chum (*Oncorhynchus keta*). A total of 60 whole fish from each species were first weighed, then placed in a light box to take their picture. Weight compared with view area as well as length and width correlations were developed. In addition the effect of “hump” development (see text) of pink salmon on this correlation was investigated. It was possible to predict the weight of a salmon by view area, regardless of species, and regardless of the development of a hump for pinks. Within pink salmon there was a small but insignificant difference between predictive equations for the weight of “regular” fish and “humpy” fish. Machine vision can accurately predict the weight of whole salmon for sorting.

Keywords: image processing, salmon, size, sorting

Introduction

In 2008, salmon was ranked 3rd by value (\$395 million) in the U.S. domestic landings behind crabs and shrimp (NMFS 2009). This constitutes a 3% increase in salmon value (\$13.3 million) despite a 26% decrease in harvest, compared to 2007. Alaska's share in the U.S. salmon landings was 97%. Total of 5 species of salmon are harvested in Alaska: Chinook or king (*Oncorhynchus tshawytscha*) constituted less than 1.5% of the catch in 2008 but 8% of the total value of salmon; chum or dog salmon (*Oncorhynchus keta*) catch in 2008 was 19% of total and constituted 17% of salmon value; coho or silver salmon (*Oncorhynchus kisutch*) was about 6% of the catch and 11% of salmon value; pink or humpback salmon (*Oncorhynchus gorbuscha*) was about 40% of the catch with 19% of the value; sockeye or red salmon (*Oncorhynchus nerka*) was 34% of the catch and 44% of the value.

After delivery into a processing plant, salmon are sorted by species, by size, and by the degree of skin watermarking which occurs when physiological changes occur after the fish start heading towards freshwater. Sorting is typically done by a number of people standing shoulder to shoulder by a conveyor belt or a chute, to sort the fish by hand. Sometimes a scale is used to quickly check the weight of the fish. Some species of salmon, especially when in the “ocean colors,” that is, bright silver, are difficult to distinguish from other species. Alaska Seafood Marketing Inst. (ASMI) has a Color Evaluation Guide for Pacific Salmon especially for the degree of skin watermarking (ASMI 2004); however, the large number of ASMI watermarking grades and the large natural variability cause uncertainties during sorting by watermarking.

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Machine vision, that is, real-time capture of images with digital cameras, analysis of these pictures, and decision-making based on its result, has been applied to sort many food and agricultural materials (Balaban and others 1994; Singh and Delwiche 1994; Watkins and others 2000; Lorestani and others 2006). The advantage of this approach is its objectivity, its ease in keeping records, and its quantitative capabilities. Fish of different species could be differentiated by computer vision, and there was a good correlation between the view area in pixels and mass of 3 species of fish (Zion and others 1999). Strachan (1990) and Strachan and others (1990) used computer vision to recognize fish species by shape analysis for sorting and size-grading. Analysis by machine vision of Atlantic salmon (*Salmo salar*) fillet color (Misimi and others 2007), shape and size (Misimi and others 2008a, 2008b), and effects of rigor, ice, and perimortem stress on properties (Erikson and Misimi 2008) have been reported. Pacific salmon research in this area is scarce. Since Pacific salmon caught in Alaska has an intense and short season, any effort in automation of the handling and processing would be of economic value.

The objective of this study was to develop weight compared with view area correlations for 4 species of Alaska salmon with the aim of automating sorting by size. The effect of “hump” development by male pink salmon on these correlations was also investigated.

Materials and Methods

Fish samples

Four Pacific salmon species were used: pink or humpback (*Oncorhynchus gorbuscha*), red or sockeye (*Oncorhynchus nerka*), silver or Coho (*Oncorhynchus kisutch*), and chum or dog (*Oncorhynchus keta*). Sixty whole fish from each species were randomly taken from a feeder chute in a processing plant in Kodiak, Alaska on August 18 and 19, 2009 as they were delivered from the fishing boat to the plant. The species designations were judged by the plant personnel. On August 25 another set of pictures were taken with the aim of imaging watermarked (and “humpback” in the case of males

who undergo transformation upon exposure to fresh water) pink salmon.

Weighing and imaging

A scale with a maximum load of 100 kg and an accuracy of 50 g was used to weigh each fish before taking its picture (Check-Weigh, Measurement Systems Inc., Seattle, Wash., U.S.A.; model MSI 6200). This is a scale typically used in processing plants. The fish was placed in a tared plastic tub before weighing, and the scale was “zeroed” for every fish.

A light box, designed and built by the researchers, similar to described by Luzuriaga and others (1997), but with the dimensions 124.5-cm high, 66-cm deep, and 86-cm wide was placed in the plant, close to the chute. Fluorescent bulbs providing D65 illumination were used behind a translucent Polycast acrylic (nr 2447, Polycast Technology Corp., Stanford, Conn., U.S.A.) 6.35 mm thick with 51% light transmission. The door of the light box was closed during picture-taking to eliminate interference from ambient light.

The fish was placed in the light box, under a Nikon D200 SLR digital camera (Nikon Corp., Tokyo) with a Nikon AF-S 18-200 mm F/3.5-5.6G ED-IF AF DX VR lens. All adjustments of the camera were set to “manual” (Table 1) for repeatability of pictures. The Nikon WT-3 wireless accessory was used to control the camera remotely from the computer, and to transfer the images to the computer immediately after acquisition, with the help of the Nikon Camera Control Pro software. The camera was mounted on a Bogen-Manfrotto Maxi Repro Stand Lite assembly (Bogen Imaging Inc., Ramsey, N.J., U.S.A.) with the lens looking downward. The setup was placed inside the light box.

A reference square of 5-cm × 5-cm was carefully cut from a dark matte carton. The square’s color was measured 3 times by a Minolta CR-300 (Tokyo, Japan) hand-held colorimeter after calibration, and the average was used as the color of the square. The square was used as a reference for both size and color, to calibrate image properties during analysis, and was included in every picture by placing it inside the light box.

Image analysis

The jpg images from the Nikon camera were entered into the Corel PhotoPaint (Corel Corp., Ottawa, Ontario, Canada) to reduce the size of the images to 900 × 602 pixels. The LensEye software (Engineering & Cyber Solutions, Gainesville, Fla., U.S.A.) was used to analyze the image:

Table 1 – Camera settings for the Nikon D200 camera.

Setting	Specification
Device	Nikon D 200
Lens	VR-18-200 mm F 3.5-5.6 G
Sensitivity	ISO 200
Optimized images	Custom
High ISO NR	Off
Exposure mode	Manual
Metering mode	Multi-pattern
Shutter speed and aperture	1.3 s and f/10
Exposure compensation (in camera)	0 EV
Focus mode	AF-S
Long exposure NR	Off
Exposure compensation (by capture NX)	0 EV
Sharpening	None
Tone compensation	Normal
Color mode	Mode I
Saturation	Normal
Hue adjustment	0
White balance	Direct sunlight

- (i) The view area of the fish was calculated by converting the number of pixels to centimeter squared. Since the surface area of the reference square was 25 cm²:

$$\text{Fish view area (cm}^2\text{)} = \text{Fish pixels/reference square pixels} \times 25 \quad (1)$$

- (ii) The perimeter pixels of the fish were found. The perimeter was approximated to a polygon by reducing the perimeter points to 100 evenly spaced points.
 - (a) Starting with an angle of 0 degrees, a line was drawn above the fish.
 - (b) The distances from the line to all of the points of the polygon were calculated
 - (c) The smallest distance was taken, and the line was translated to that point
 - (d) Another line, perpendicular to the above line was drawn left of the fish
 - (e) The distances from this line to all of the points of the polygon were calculated
 - (f) The smallest distance was taken, and the line was translated to that point
 - (g) The line below (parallel to the line above), and the line at the right (parallel to the line at the left) were also processed similarly.
 - (h) The area of the rectangle was calculated
 - (i) The angle was changed by one degree, and the process was repeated.
 - (j) The rectangle with the smallest area was taken as the “best fitting rectangle.”
- (iii) The length and the width of this rectangle were also used in correlations to weight estimation (Figure 1).

Regression analysis

The following correlations between weight, view area and rectangle dimensions were tried:

$$\text{Linear: } Y = A + BX_1 \quad (2)$$

$$\text{Linearized: } Y = A \exp(BX_1) \quad (3)$$

$$\text{Linearized: } Y = A + B \ln(X_1) \quad (4)$$

$$\text{Linearized: } Y = AX_1^B \quad (5)$$

$$\text{Polynomial: } Y = C_0 + C_1 X_1 + C_2 X_1^2 \quad (6)$$

$$\text{Surface 1: } Y = C_0 + C_1 X_1 + C_2 X_2 + C_3 X_3 \quad (7)$$

$$\text{Surface 2: } Y = C_0 + C_1 X_1 + C_2 X_2 + C_3 X_3 + C_4 X_1^2 + C_5 X_2^2 + C_6 X_3^2 \quad (8)$$

In the equations, Y = weight (kg), X_1 = view area (cm²), X_2 = enclosing rectangle length (cm), X_3 = enclosing rectangle width (cm), A , B , C_i are coefficients.

Humpback fish

Male pink salmon develop a “hump” during the later stages of their migration towards fresh water (Figure 2). This changes the shape of the fish and may affect the weight compared with view area correlation. Therefore, images of fish that developed an obvious hump were separated from the rest of the pink salmon, and

both the “humpback” and “regular” pink salmon data were also analyzed separately.

Results and Discussion

The results of fitting Eq. (2) to (8) to each species of salmon, and to all the fish considered together are shown in Table 2 and 3. In Table 2 the R^2 values range between 0.928 and 0.987 for simple fits (2 coefficients). In general, Eq. (5) (power curve: $Y = AX^B$) resulted in the highest R^2 value. This is logical, since the weight (which is related to volume, assuming the density of the fish is constant) and the view area (a surface) are bound with a dimensional relation such as:

$$\text{Volume dimension} = (\text{Area dimension})^{3/2} \quad (9)$$

It can be seen that in Table 2 the B values in the last equation for all correlations are close to 1.5. Considering that the tail and fins contribute much to the area, and not much to the weight, some deviation from ideal is to be expected. It is also possible that the density of the tail and fins are different than that of the rest of the fish, although this was not tested. It should also be noted that the R^2 values of the pink salmon were the lowest within all species, regardless

of the equation used in Table 2 and 3. This may be due to the hump development of the pink salmon males, and this may “scatter” the data, as will be discussed subsequently.

In Table 3, the fitting results for equations with more coefficients are presented. It is expected that in general an increase in the number of coefficients will increase the R^2 value of the fit. This is generally the case for individual species and combined fish; however, the highest R^2 in both tables is for the power curve fitted to all salmon (last column) in Table 2. This implies that a simple equation using only the view area of the fish may be sufficient to satisfactorily estimate the weight of any salmon belonging to the 4 species used. Therefore, the extra work (and therefore computing time) necessary to draw the best bounding box to calculate its length and width may not be necessary.

The graphs of all the fits for all the species will not be shown for brevity. Since pink salmon represents the most scattered data, with a potential complication of humpy males after they undergo transformation upon entering freshwater, its curve fits will be displayed. In Figure 3, the simple 2-coefficient equation fits are displayed for pink salmon. It can be seen that the exponential (Eq. 3) and logarithmic (Eq. 4) fits deviate at both end of the data range. The linear (Eq. 2) and power (Eq. 5) curves are close, also evidenced

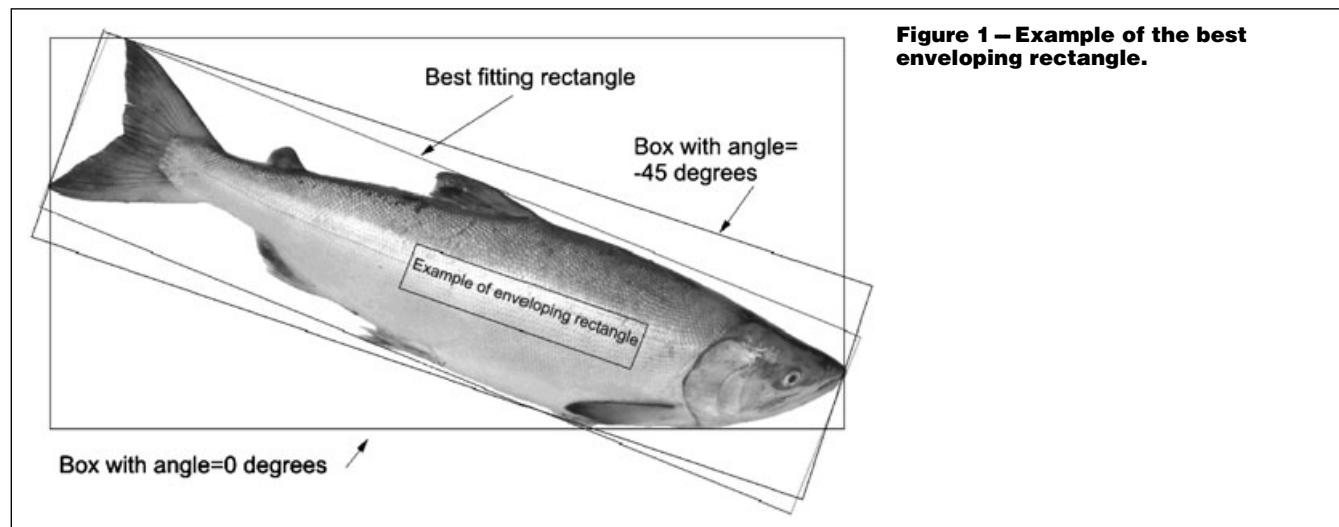


Figure 1 – Example of the best enveloping rectangle.



Figure 2 – Example of pink salmon with obvious hump (top), and regular pink salmon (bottom).

by their similar R^2 values in Table 2. The error range of the scale (± 50 g) is shown as dashed lines around the linear fit. Figure 4 shows fits of equations with multiple coefficients to the pink salmon data. There is marginal improvement in the fit, as evidenced by the small increase in the R^2 values between Table 2 and 3 for pink salmon column. Again the weighing error is indicated by the dashed lines around the polynomial fit. In the case of pink salmon there is very little difference between the 3 fits.

Figure 5 presents separate linear fits for “regular” and “humpy” pink salmon. For this, the data were manually separated between the 2 types of fish. It is apparent that for a given weight (horizontal line), the “humpy” fish generally fall on the higher view area side (right). Therefore, there is a discernible difference in the weight compared with view area relationship in the case of “humpy” pink salmon. Table 4 shows the statistical analysis of the 2 regression lines fit to both the humpy pinks and the regular pinks shown in Figure 5. Comparing the 95% confidence intervals of the intercepts (-0.9396 to -0.2103 for humpies, -0.80675 to -0.51296 for regu-

lars) and slopes (0.0367 to 0.00505 for humpies, 0.00446 to 0.00513 for regulars) both intervals overlap. Therefore there is no statistical difference between the weight compared with view area correlations between the 2 groups.

The plot of linear and power curve fits for all species combined are shown in Figure 6. The other simple fits are not shown for simplicity. It is apparent that the power curve is a better model for this correlation, also as evidenced by its slightly higher R^2 value. It seems to follow the experimental data better at the extremes of the data range. The black circles represent the “humpy” pink salmon. Although in every case these data points are to the right (that is, bigger view area) of the fitted equation, in the overall scheme they are not very different from the natural variation in the other fish.

In Figure 7, the fit of the multi-coefficient equations to all of the salmon data is shown. There is no difference in the R^2 values between the 2nd-order polynomial (Eq. 6) and the quadratic surface (Eq. 8) depicted as “surface 2” in Figure 7. Therefore, the simpler

Table 2— Simple regression results for the correlation of view area to weight for 4 species of salmon and for all salmon combined. Y = weight (kg), X = view area (cm²), A and B = coefficients.

	Pink	Red	Silver	Chum	All 4 species
Equation = $Y = A + BX$					
A	-0.6678	-0.8568	-1.1109	-1.0546	-1.0321
B	0.0048	0.00524	0.005771	0.00567	0.00561
R^2	0.945	0.970	0.976	0.970	0.983
Equation = $Y = A \text{ Exp}(BX)$					
A	0.3112	0.5859	0.5914	0.5876	0.48636
B	0.00345	0.00224	0.0022	0.002257	0.00251
R^2	0.928	0.947	0.969	0.951	0.966
Equation = $Y = A + B \ln(X)$					
A	-11.104	-18.142	-20.4564	-20.3483	-18.1274
B	2.0643	3.1985	3.5807	3.5567	3.2135
R^2	0.937	0.961	0.937	0.955	0.953
Equation = $Y = AX^B$					
A	0.000153	0.000315	0.000267	0.000224	0.000183
B	1.503	1.3889	1.4179	1.4451	1.4751
R^2	0.948	0.968	0.983	0.976	0.987

Table 3— Polynomial and multivariable fits to 4 species of salmon and all fish combined

Equation 1: $Y = C_0 + C_1X_1 + C_2X_1^2$

Equation 2: $Y = C_0 + C_1X_1 + C_2X_2 + C_3X_3$

Equation 3: $Y = C_0 + C_1X_1 + C_2X_2 + C_3X_3 + C_4X_1^2 + C_5X_2^2 + C_6X_3^2$

Y = weight (kg); X_1 = view area (cm²); X_2 = Bounding box length (cm); X_3 = Bounding box width (cm); C_i = coefficients.

	Pink	Red	Silver	Chum	All 4 species
Eq. 1					
C_0	-0.715	-0.9	-0.1681	-1.07804	-0.4341
C_1	0.0050	0.0054	0.00281	0.005744	0.003545
C_2	-2.17×10^{-7}	-1.12×10^{-7}	2.22×10^{-6}	-5.61×10^{-8}	1.69×10^{-6}
R^2	0.945	0.970	0.981	0.970	0.985
Eq. 2					
C_0	-0.13988	-0.5193	-0.1644	-0.2697	-0.68303
C_1	0.005829	0.005685	0.007253	0.00669	0.006208
C_2	-0.01415	-0.0099	-0.0175	-0.02176	-0.00874
C_3	-0.01976	1.35×10^{-5}	-0.0537	-0.00156	-0.01208
R^2	0.947	0.971	0.978	0.974	0.983
Eq. 3					
C_0	5.8027	-12.8786	4.6983	-1.2432	-0.95824
C_1	0.014746	-0.0053	0.008145	0.0062	0.00341
C_2	-0.27104	0.4695	-0.15	-0.0273	0.03468
C_3	-0.2095	0.1543	-0.23175	0.17205	-0.06189
C_4	-9.75×10^{-6}	9.3×10^{-6}	-1.23×10^{-6}	3.09×10^{-7}	1.89×10^{-6}
C_5	0.00244	-0.00394	0.001175	2.33×10^{-5}	-0.00028
C_6	0.007405	-0.00574	0.006194	-0.00518	0.001708
R^2	0.947	0.977	0.982	0.975	0.985

Table 4 – Statistical analysis of the regression fit to pink salmon, regular, and humpy.

	Coefficients	Standard error	t Stat	P-value	Lower 95%	Upper 95%
Humpy pinks						
Intercept	−0.57494	0.17536	−3.27866	0.003584	−0.9396	−0.21026
Slope	0.00436	0.00033	13.14388	1.3310 ^{−11}	0.00367	0.00505
Regular pinks						
Intercept	−0.65985	0.07317	−9.01825	3.8710 ^{−12}	−0.80675	−0.51296
Slope	0.004793	0.00017	28.57947	4.710 ^{−33}	0.004456	0.005129

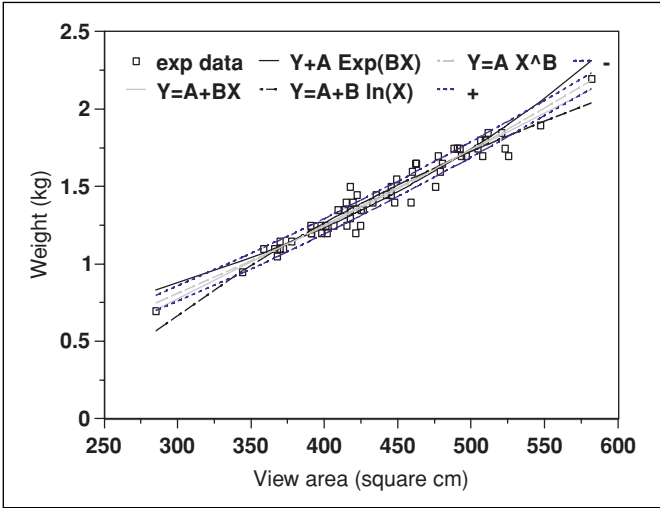


Figure 3—Different simple equation fits to weight compared with view area for pink salmon (all fish including humpies were used).

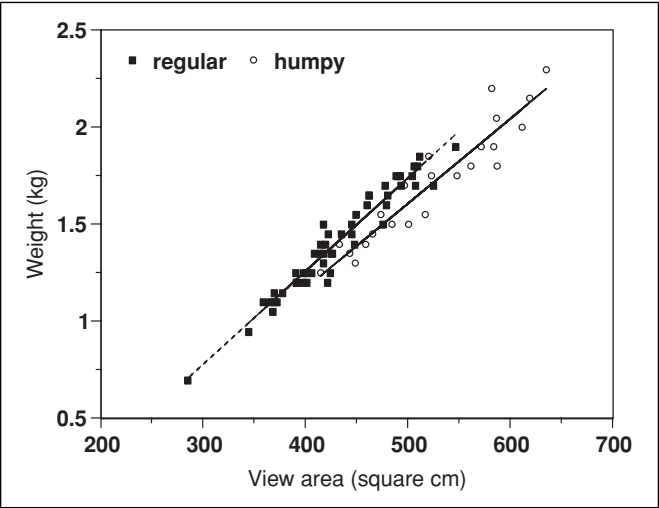


Figure 5—Plot of linear fit of weight compared with view area of regular and humpy pink salmon.

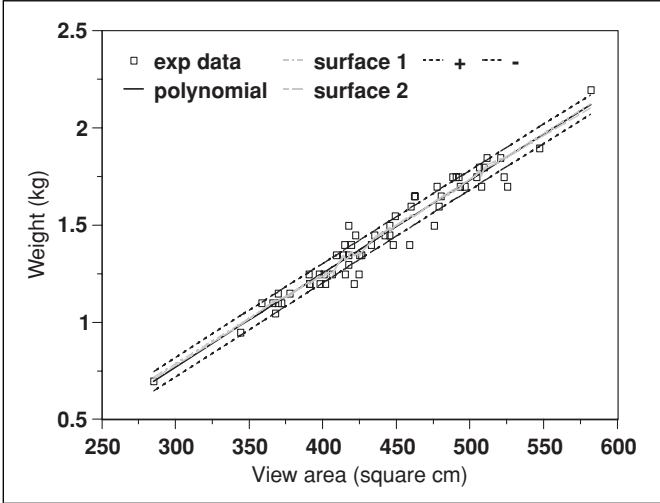


Figure 4—Different equation fits to weight compared with view area for pink salmon (all fish including humpies were used). Dashed lines show the accuracy of the scale used.

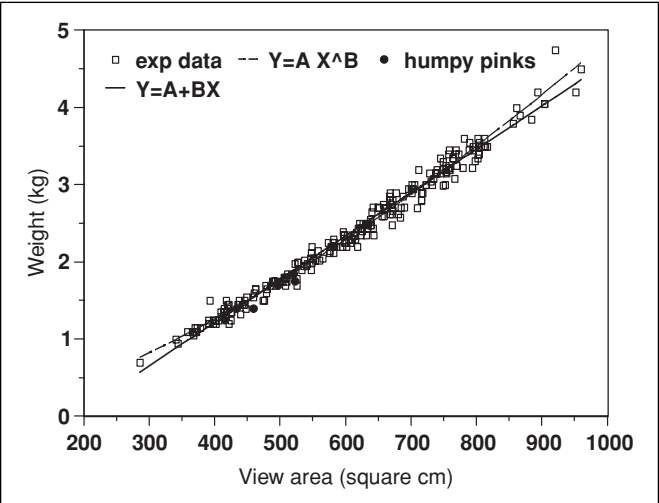


Figure 6—Different simple equation fits to weight compared with view area for all salmon (all fish including humpies were used). The black circles represent the humpy pink salmon.

approach is to use the 2nd-order polynomial and eliminate the need for the calculation of the best enveloping rectangle.

Conclusions

The weight of any 1 of the 4 species of salmon studied in this study can be estimated accurately by measuring its view area by machine vision, and then using either a power curve, or a 2nd-order polynomial to calculate its weight. This is a simple method that does not depend on the orientation or curvature of the fish. This allows objectively sorting of these salmon species by weight, using machine vision.

For possibly better accuracy, the effects of the fins (sometimes positively contributing to the view area, sometimes not, for the same weight of the fish) and the tail (sometimes fully expanded, sometimes folded) may be removed from the correlation. This requires the development of software to further process the salmon image to “erase” the tail and the fins sticking out from the body. The added computation time should be compared with the increase in the prediction accuracy, and whether the desired accuracy can be attained without this additional process.

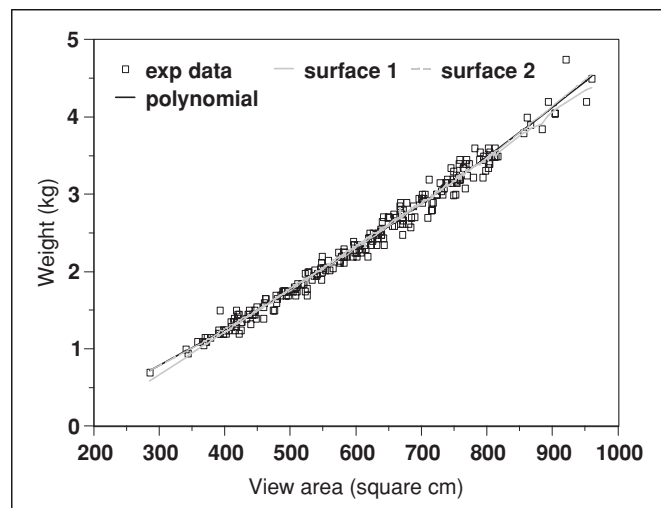


Figure 7 – Different equation fits to weight compared with view area for all salmon (all fish including humpy pinks were used).

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